

Comparing Quantities

This week's big question: what two quantities am I comparing? Ratios, rates, and proportions are all different ways of answering it.

What Does a Ratio Compare?

A ratio compares two quantities. Some ratios compare a part to another part — 5 forwards to 7 defenders. Others compare a part to a whole — forwards to the entire 12-player team. These aren't the same statement: 5 forwards out of 12 total players is $\frac{5}{12}$ of the team, not $\frac{5}{7}$.

Writing and Simplifying Ratios

The same ratio can be written three ways: 3 to 2, 3:2, or $\frac{3}{2}$. Order matters — 3:2 and 2:3 describe different comparisons. Simplify a ratio the same way you'd simplify a fraction, using the greatest common factor: 18:12 simplifies to 3:2. The relationship being compared doesn't change, just the numbers used to describe it.

Equivalent Ratios

Scale a ratio up or down by multiplying or dividing *both* quantities by the same factor.

Common mistake: if a recipe ratio needs more sugar, some students add the same amount to every ingredient instead of multiplying. Equivalent ratios come from multiplying or dividing, not adding.

What Is a Rate?

A rate is a ratio that compares quantities with *different* units — miles to hours, dollars to items. Every rate is a ratio, but not every ratio is a rate. Keep the units attached as you work; they're part of the answer, not an afterthought.

Unit Rates

A unit rate divides so the second quantity becomes exactly 1 — dollars per one item, miles per one hour.

\$104 for 8 hours of work: $104 \div 8 = \$13$ per hour.

A unit rate isn't automatically "bigger number $\div$ smaller number." It's whichever quantity the question calls the reference — the "per one" unit.

Comparing Rates: Best Buy

To compare two offers, calculate each one's unit rate, then compare those directly.

A higher total price doesn't automatically mean a worse deal — the unit rate decides value, not the sticker price.

Proportions

A proportion is a statement that two ratios (or rates) are equal — for example, $4/5 = x/30$.

Solving Proportions

Two paths to the same answer: cross multiply, or find a unit rate and scale. Either way, the proportion has to be set up with matching quantities in matching positions first — cross multiplication only gives a correct answer when that's true.

Solve $4/5 = x/30$: cross multiply: $4 \times 30 = 5 \times x$, so $120 = 5x$, $x = 24$.

Unit Conversion

A conversion factor is just a ratio, arranged so the unit you don't want cancels out. Flip it, and it cancels the wrong thing — watch for a result that's obviously the wrong size.

Does My Answer Make Sense?

Check	Ask
Size	Is the answer approximately the size I expected?
Direction	Did I scale up or down the way the situation actually requires?
Units	Did the units cancel or combine the way they should?
Context	Does this answer make sense in the real situation?

Error Detective

Knowledge — “I don't remember how to set up a proportion.”

Fix: review the setup rule — matching quantities in matching positions.

Process — correct setup, wrong cross multiplication

The proportion was set up correctly but the cross-multiplication steps went wrong. Fix: review the steps and retry.

Execution — correct setup and cross multiplication, arithmetic slip

Everything about the method was right; the final division had a small mistake. Fix: check the calculation and verify with estimation.

Comprehension — comparing forwards to defenders instead of forwards to the whole team

The question asked for a part-to-whole comparison; the student compared part-to-part instead. Fix: reread the question and identify exactly what's being compared.

Strategy — comparing total prices instead of unit rates

The reasoning wasn't wrong, just inefficient and easy to misread. Fix: convert to a unit rate before comparing.

Quick Reference

Ratio: a comparison of two quantities.

Rate: a ratio comparing different units.

Unit rate: a rate where the second quantity equals one.

Proportion: a statement that two ratios or rates are equal.

Solving a proportion: cross multiply, or find the unit rate and scale.

Check your answer: Size → Direction → Units → Context.

Keep this page. You'll use it all year.